

INTEGRAZIONE PER SOSTITUZIONE

I, J INTERVALLO, $f: I \rightarrow \mathbb{R}$, $g: J \rightarrow \mathbb{R}$ T.C.

- $\text{Im } g \subseteq I$

- f CONTINUA

- g DERIVATA, g' CONTINUA

ALLORA

$$\int f(g(x))g'(x)dx = \underbrace{\int \overbrace{f(t)}^{F'(t)} dt}_{F(g(x))} \Big|_{t=g(x)}$$

PERCHÉ?

• $F(g(x))$ È UNA PRIMITIVA DI $f(g(x))g'(x)$

es) $\int x \sin(x^2) dx$

$$= \int \frac{1}{2} \sin(x^2) 2x dx$$

$$= \frac{1}{2} \int \sin(x^2) \underbrace{2x}_{g'(x)} dx$$

$$= \frac{1}{2} \int f(t) dt \Big|_{t=g(x)}$$

$$= \frac{1}{2} \int \sin t dt \Big|_{t=x^2}$$

$$= \frac{1}{2} (-\cos t) \Big|_{t=x^2} + C$$

$$= -\frac{1}{2} \cos(x^2) + C$$

OPPURE

$$\int x \sin x^2 dx$$

$$x^2 = t \quad 2x dx = dt \quad x dx = \frac{dt}{2}$$

$$= \int \sin t \frac{dt}{2} = \frac{1}{2} \int \sin t dt$$

$$= -\frac{1}{2} \cos t + C = -\frac{1}{2} \cos(x^2) + C$$

• INTEGRALI DI FUNZIONI ELEMENTARI

$$\cdot \int \log x dx$$

$$= \int 1 \cdot \log x dx$$

$$\begin{array}{cc} \cancel{f(x)} & \cancel{g'(x)} \\ g'(x) & f(x) \end{array}$$

$$f = \log x \quad f' = \frac{1}{x}$$

$$g' = 1 \quad g = x$$

$$= f(x)g(x) - \int g(x)f'(x) dx$$

$$= \log x \cdot x - \int x \cdot \frac{1}{x} dx$$

$$= x \log x - \int 1 dx$$

$$= x \log x - x + C$$

$$b) \int \arctan x \, dx$$

$$\int \underbrace{1}_{g'} \cdot \underbrace{\arctan x}_f$$

$$g' = 1 \quad g = x$$

$$f = \arctan x \quad f' = \frac{1}{1+x^2}$$

$$= x \cdot \arctan x - \int x \cdot \frac{1}{1+x^2} dx$$

$$t = x^2 + 1$$

$$2x \, dx = dt$$

$$x \, dx = \frac{dt}{2}$$

$$= x \cdot \arctan x - \int \frac{1}{t} \frac{dt}{2}$$

$$= x \cdot \arctan x - \frac{1}{2} \log |t| + C$$

$$= x \cdot \arctan x - \frac{1}{2} \log \log (x^2 + 1) + C$$

• INTEGRAZIONE DI FUNZIONI RAZIONALI

$$\int \frac{N(x)}{D(x)} dx$$

$N(x), D(x)$: POLINOMI

$$\int \frac{1}{x^2+3} dx = \int \frac{1}{3\left(\frac{x^2}{3}+1\right)} dx$$

$$= \frac{1}{3} \int \frac{1}{\frac{x^2}{3}+1} dx$$

$$= \frac{1}{3} \int \frac{1}{\underbrace{\left(\frac{x}{\sqrt{3}}\right)^2 + 1}} \underbrace{dx}$$

$$\frac{x}{\sqrt{3}} = t \quad \frac{1}{\sqrt{3}} dx = dt \quad dx = \sqrt{3} dt$$

$$= \frac{1}{3} \int \frac{1}{t^2 + 1} \sqrt{3} dt$$

$$= \frac{\sqrt{3}}{3} \int \frac{1}{t^2 + 1} dt$$

$$= \frac{\sqrt{3}}{3} \arctan t + C$$

$$= \frac{\sqrt{3}}{3} \arctan\left(\frac{x}{\sqrt{3}}\right) + C$$

$$\bullet \int \frac{x^2 - 1}{x + 2} dx$$

$$\begin{array}{l} \text{GRADO } N(x) = 2 \\ \text{GRADO } D(x) = 1 \end{array} \quad \& \quad 2 \geq 1$$

SE GRADO $N(x) \geq$ GRADO $D(x)$:

$$\frac{44}{7} = 6 + \frac{2}{7} \rightarrow \text{RESTO} \quad R < 7$$

$$\frac{N(x)}{D(x)} = \underbrace{Q(x)}_{\text{GRADO } Q} + \frac{R(x)}{D(x)} \quad \begin{array}{l} \text{GRADO } R(x) \\ \uparrow \\ \text{GRADO } D(x) \end{array}$$

$= \text{GRADO } N - \text{GRADO } D$

$$\frac{x^2 - 1}{x + 2} = Q(x) + \frac{R(x)}{x + 2}$$

$$\text{GRADO } Q = 2 - 1 = 1$$

$$\text{GRADO } R < 1 \Rightarrow \text{GRADO } R = 0$$

$$\frac{x^2 - 1}{x + 2} = \boxed{Ax + B} + \frac{\boxed{C}}{x + 2}$$

$$\frac{x^2 - 1}{x + 2} = \frac{(Ax + B)(x + 2) + C}{x + 2} =$$

$$= x^2 - 1 = Ax^2 + 2Ax + 6x + 2B + C$$

$$= x^2 - 1 = Ax^2 + (2A + 6)x + (2B + C)$$

$$= \begin{cases} A = 1 \\ 2A + B = 0 \\ -1 = 2B + C \end{cases} \Leftrightarrow \begin{cases} A = 1 \\ B = -2A = -2 \\ C = -1 - 2B = 1 + 4 = 3 \end{cases}$$

$$\frac{x^2 - 1}{x + 2} = x - 2 + \frac{3}{x + 2}$$

$$\int \frac{x^2-1}{x+2} dx = \int x-2 + \frac{3}{x+2} dx$$

$$= \int x dx - \int 2 dx + 3 \int \frac{1}{x+2} dx$$

$$= \frac{x^2}{2} - 2x + 3 \log|x+2| + C$$

OPPURE (PER TROVARE A, B, C)

$$\int \frac{2x^2+x}{x-1} dx$$

$$\frac{2x^2+x}{x-1} = Ax + B + \frac{C}{x-1}$$

$$= 2x + 3 + \frac{3}{x-1}$$

$$\begin{array}{r|l} 2x^2+x & x-1 \\ 2x^2-2x & 2x+3 \end{array} \rightarrow Q$$

$$\emptyset \quad 3x$$

$$3x-3$$

$$\emptyset \quad (3) \rightarrow \text{RESTO}$$

OPPURE (ANCORA PIÙ VELOCE)

$$\frac{2x^2 + x}{x - 1} = \frac{2x^2 - 2x + 2x + x}{x - 1}$$

$$= \frac{2x^2 - 2x}{x - 1} + \frac{3x}{x - 1}$$

$$= 2x + \frac{3x - 3 + 3}{x - 1}$$

$$= 2x + \frac{3x - 3}{x - 1} + \frac{3}{x - 1} = 2x + 3 + \frac{3}{x - 1}$$